NeuroScan AI — Multimodal Stroke Screening Report

GMU MASE 2026 AI Healthcare Competition | Model Version: 1.0.0-MASE2026 | Generated: 2026-11-02 09:14:22 UTC

IMPORTANT CLINICAL & REGULATORY LIMITATION: NeuroScan AI is an experimental research and educational software prototype developed for the GMU MASE 2026 competition. It is NOT a certified or clinically validated medical device and must never be used to independently diagnose, treat, or rule out stroke or any neurological pathology. 1) Any sudden neurological symptom (Facial droop, Arm weakness, Speech abnormality) must trigger immediate emergency medical escalation regardless of AI output. 2) Non-contrast CT (NCCT) sensitivity for hyperacute ischaemic stroke is limited; a negative CT haemorrhage result NEVER rules out ischaemic stroke. 3) Normal vital signs (BP, HR, SpO2) never exclude acute cerebrovascular events.

Assessment ID:	ASM-DEMOB002	**Patient Ref ID:**	DEMO-SCENARIO-B
Patient Age:	61	**Symptom Onset:**	90 minutes ago
Record Provenance:	SIMULATED_DEMO RECORD	**Assessment Timestamp:**	2026-11-02T09:14:22

1. Integrated Urgency Assessment & Clinical Recommendation

URGENCY CLASSIFICATION: **IMMEDIATE_EMERGENCY_ESCALATION**

RECOMMENDED NEXT ACTION: IMMEDIATE EMERGENCY ESCALATION: Activate emergency stroke protocol (Call Emergency Services / Stroke Team) immediately. Do not delay care for additional uploads. Note: Negative CT haemorrhage never excludes ischaemic stroke.

2. FAST Neurological Screening & Vital Signs

FAST Domain	Manual Observation	Experimental AI Status	Vital Sign	Recorded Value & Provenance
Facial Weakness (F)	YES	ASYMMETRY_FLAGGED (score=0.24)	Blood Pressure	134/82 mmHg [SIMULATED_DEMO]
Arm Weakness (A)	YES	DRIFT_FLAGGED	Heart Rate	88 bpm
Speech Difficulty (S)	YES	See Acoustic Module	SpO2 / Temp	SpO2: 98.0% | Temp: 36.8 C

3. AI Brain CT Scan Analysis (Research Decision Support — Not a Diagnosis)

Haemorrhage Subtype	Model Score	Interpretation Note
Intraparenchymal	3.0%	Below 50% Threshold
Intraventricular	1.0%	Below 50% Threshold
Subarachnoid	2.0%	Below 50% Threshold
Subdural	4.0%	Below 50% Threshold
Epidural	1.0%	Below 50% Threshold
Any Haemorrhage	6.0%	Below 50% Threshold
Model Uncertainty (Entropy)	0.14	Weights Status: DEMO_REFERENCE_RECORD
Experimental Ischaemic Burden	3.8%	Status: SUSPECTED_EARLY_MCA_TERRITORY

4. Multimodal Synthesis, Safety Escalations & Limitations

Reasons for Escalation:	Sudden facial weakness reported during FAST screening.; Sudden arm weakness/drift reported during FAST screening.; Sudden speech impairment reported during FAST screening.; Experimental AI facial landmark screening flagged labial-canthal asymmetry.; Experimental AI pose tracking flagged unilateral arm drift.; MANDATORY SAFETY RULE: Positive stroke symptoms with negative CT haemorrhage require immediate emergency evaluation for suspected acute ischaemic stroke.
Cross-Modal Conflicts:	CRITICAL CROSS-MODAL DISCORDANCE: Clinical FAST screening is POSITIVE while CT Haemorrhage classification is NEGATIVE. A negative haemorrhage CT NEVER rules out acute ischaemic stroke.
Missing Information:	Acoustic Speech Recording Analysis

Uncertainty & Limitations:	Standard operational parameters.